

Review

Waterjet machining of biological tissues in medical surgeries: From soft tissue dissection to bone cutting

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ABSTRACT

Tissue cutting is a primary procedure in medical surgeries, during which waterjet machining is increasingly used for tissue dissection, ablation and removal because it induces the least thermal damage, enables tissue-selective separation and prevents blood loss. This paper reviews the related studies and techniques for waterjet machining in surgeries with respect to development and classification of medical waterjet, tissue types, process characteristics, tissue-waterjet interaction mechanisms, and tool solutions; and highlights the research trends for soft tissue dissection and hard tissue cutting. The review begins with the historical development and the classification of medical waterjet processes. Next, waterjet processing of soft and hard tissues is discussed, including tissue structures and properties, cutting/dissection mechanisms, process characteristics, tool design and applications. Following, waterjet solution strategies related to tissue properties, biomimetic models, tissue-process-tool interactions, waterjet mediums and devices are explored. Finally, the review highlights research gaps, challenges and future developments of waterjet techniques for low trauma, high cutting efficiency and multiple surgical applications.

1. Introduction

Cutting of biological tissues is the most frequently performed procedure in surgeries. With the rapid development of machining technologies, many advanced cutting processes have been applied to medicine. Waterjet as an emerging cutting tool has been used in surgeries to impact biological tissues with a very fine high-speed water beam formed from sterile saline for tissue deformation, separation, and dissection [1]. Compared to conventional rigid tool cutting in surgeries, which often induces excessive damage and high temperatures to cause cellular death of tissues [2], waterjet processes avoided such problems and were first used for hepatic resections in open surgeries [3]. With improved waterjet devices and surgical procedures, waterjet machining has been gradually applied in open surgeries for tissue dissection, ablation, and removal.

Table 1 reveals the applications of waterjet in surgeries. For soft tissue surgeries, the application examples of waterjet include hepatectomy, gliomas dissection in neurosurgery [9] and sweat gland resection in dermatologic surgery [10]. In recent years, laparoscopic surgery has

been used as a new surgical method for a minimally invasive way of complete resection or ablation of diseased tissues. The combination of waterjet and laparoscopic surgery has achieved good clinical results in cholecystectomy [11], dissection of tumors [12], and ablation of benign prostatic hyperplasia [13]. Further, hard tissue machining with waterjet is currently increasing its attention because the process induces the least thermal damage to bone to avoid osteonecrosis [2]. Consequently, waterjet may potentially become a useful tool for microfracture treatment [14], screw fixation in bone drilling [15], and prosthetic repair surgeries for minimally invasive removal of interface tissues around the prostheses [16]. Waterjet has also been used for needle-free injections [7] and hydrodynamic debridement [8], or as waterjet steerable needles [17]. Needle-free injections have been extensively used for vaccinations, insulin delivery, gene therapy, viable cell injections, and skin remodeling [18]. Waterjet steerable needles show promising potential for targeted cancer treatment and accurate biopsy [17]. Hydrodynamic debridement has been a newer debridement method for debriding acute and chronic wounds, such as burns and ulcers, leading to more efficient and faster heals than traditional debridement [19].

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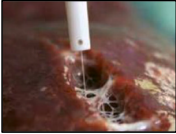
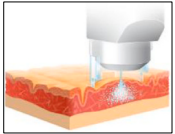
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Table 1
Waterjet applications in surgeries. All figures reprinted with permissions from Elsevier, Springer and Wiley.

Soft tissue dissections		Hard tissue cutting	Other applications	
Open surgery	Laparoscopic surgery		Needle-free injection	Hydrodynamic debridement
				
Application: liver [4]	Application: tumor [5]	In vitro research: bone cutting [6]	Application: vaccines [7]	Application: burns [8]

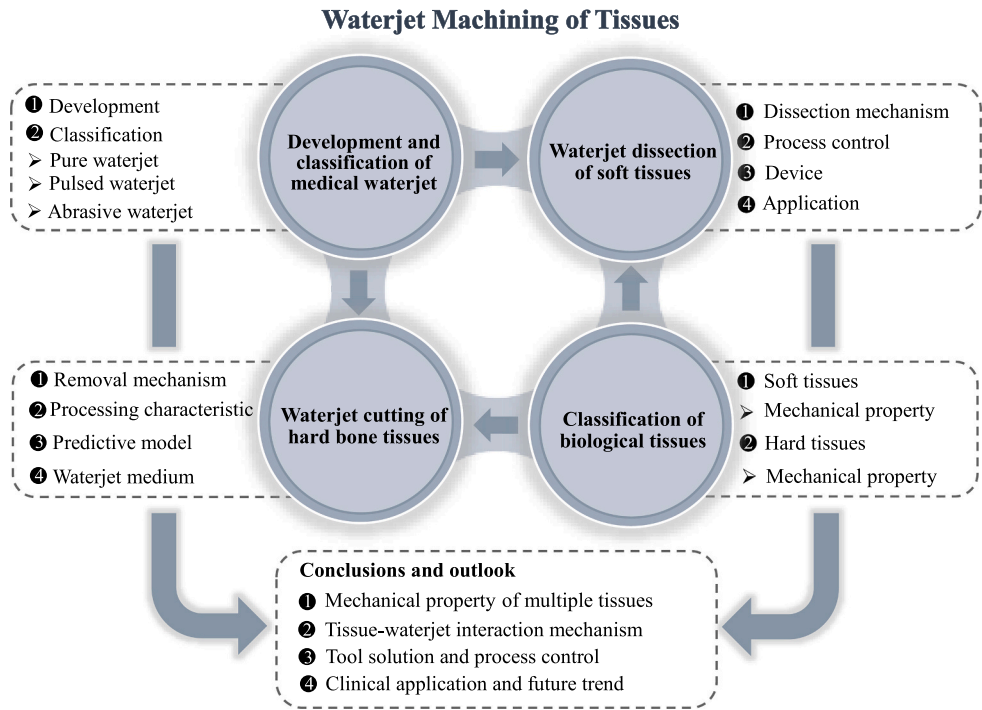


Fig. 1. Review structure.

As an emerging surgery technique, waterjet has superior advantages over traditional rigid tools to diminish excessive injury and tissue necrosis and obtain high lesion tissue removal rates with maximum protection of nerve fibers and blood vessels [20]. Thus, waterjet procedures can achieve higher complete resection rates of tumors and low intra-procedural perforation rates in tumor removal and reduce blood losses and risks of postoperative dysfunctions in high-precision neuro [21] or urology surgeries [12]. These help patients accelerate their recovery and improve their postoperative quality of life [22]. Over the past few decades, a considerable amount of research and development efforts has been made to enhance the unique waterjet advantages in medical applications. Research progress has been achieved with respect to tissue structures and properties [23], as well as waterjet process parameters [24], numerical modeling [25], tool design and applications (e.g., needle-free injection [18] and hydrodynamic debridement [19]), and clinical outcomes. Most research outcomes were tissue-specific but provided systemic comparisons for multiple waterjet processing parameters to yield cutting depth prediction models or obtain comparisons using clinical indicators (e.g., less blood, less recovery time) between waterjet and traditional devices. However, there lacks a systematic summary of the research and development of waterjet machining in medicine.

This paper aims to conduct a comprehensive review of waterjet machining of tissues in medical surgeries through extensive literature search and analysis with four objective parts shown in Fig. 1. Specifically, the development and classification of medical waterjet techniques are firstly summarized with respect to waterjet mediums, implementation methods, and water pressure evolution characteristics. Secondly, waterjet dissection of soft tissues is focused, covering various tissue structure and mechanical properties, common dissection processes, process parameters and predictive models associated with processing properties, interactions, and damage mechanisms, as well as new waterjet tool design and applications. Thirdly, waterjet cutting of hard bone tissues is summarized, including removal mechanisms, processing characteristics, and predictive models. Finally, identified research gaps are highlighted and future research prospects on waterjet for medicine are proposed.

2. The development and classification of medical waterjet techniques

As a non-thermal cutting tool, waterjet machining has been one of the focal subjects in medical manufacturing. It is of interest to understand the development and classification of medical waterjet techniques.

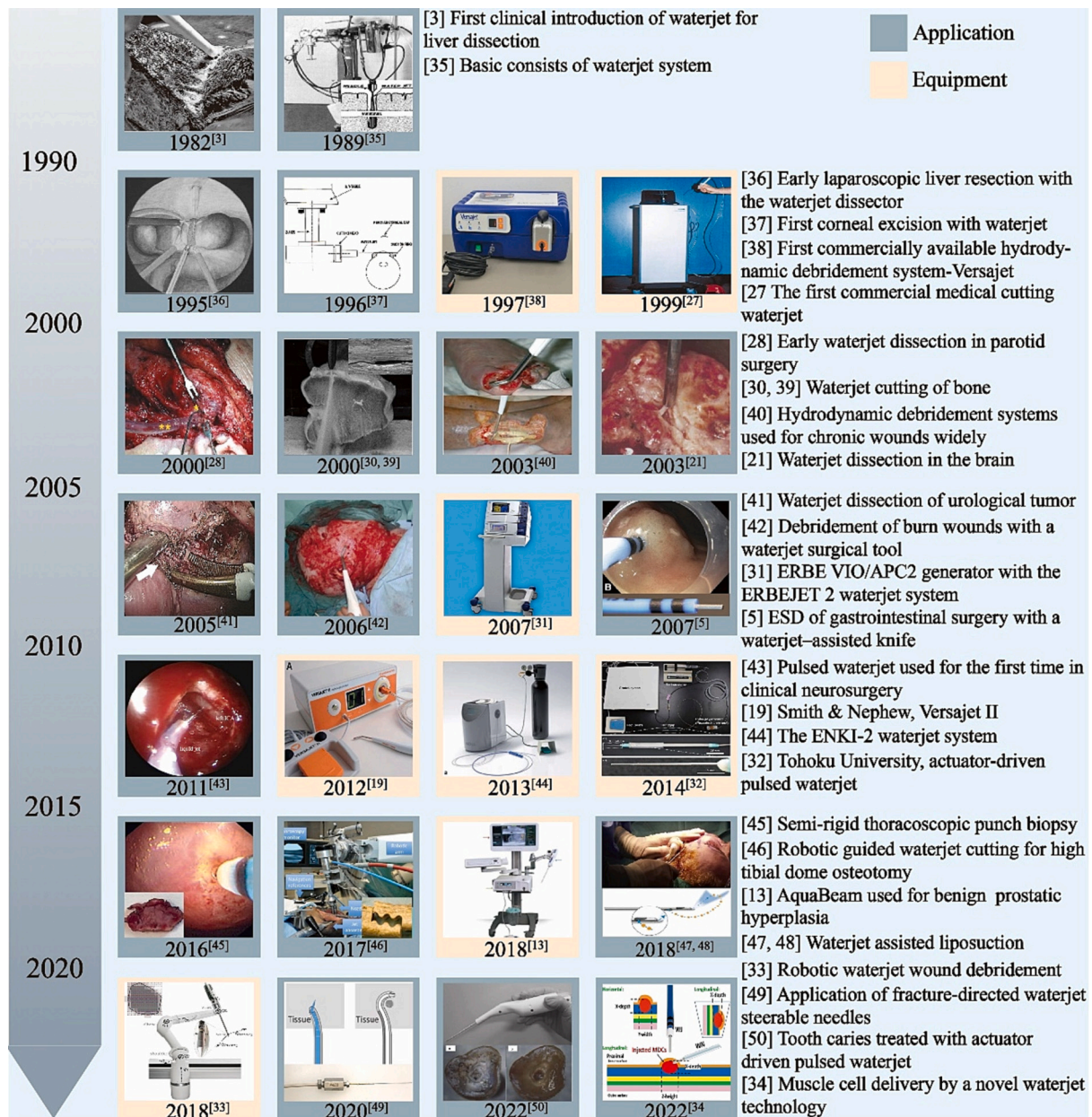


Fig. 2. Development history of waterjet in surgeries [3,5,13,19,21,27,28,30–50]. All figures reprinted with permissions from Elsevier, Springer, Wiley, Taylor & Francis, PLOS, Mary Ann Liebert, Lippincott Williams & Wilkins, De Gruyter, Assoc Research Vision Ophthalmol Ogy Inc., Oxford University Press, MA Healthcare, Georg Thieme Verlag KG, Sage Publications Inc. and Georg Thieme Verlag KG.

2.1. Development of medical waterjet

While waterjet was initially applied to industrial manufacturing (e. g., cutting and mining) dated in 1800 [26], its first medical application took place a century later. The milestones in the development of medical waterjet techniques are outlined in Fig. 2. Professor Papachristou and Barters initially introduced waterjet for liver tissue cutting in 1982 [3]. The first commercial medical waterjet device named Helix Hydro-Jet was manufactured by Erbe Elektromedizin [27] in the 1990s and gradually used in clinical surgeries, such as dissection of brain tumors [27], parotid [28], and spleen [29]. Later, the usage of waterjet was expanded to bone cutting. The first research on waterjet cutting of bone was published in the early 2000s [30]. Since then, various medical waterjet devices have also been developed, including Hybrid-Knife® [31], AquaBeam [13], and actuator-driven pulsed waterjet [32]. Over the last decade, the rapid progress in minimally invasive surgery had led

to the combination of waterjet and laparoscopic surgeries applied in dissection of tumors [12] and ablation of benign prostatic hyperplasia [13]. More recently, novel waterjet technologies have been reported, including robotic waterjet wound debridement [33], waterjet steerable needles [25], and waterjet delivery of viable cells [34]. Medical waterjet devices have been used for both open and minimally invasive laparoscopic surgeries.

2.2. Classification of medical waterjet processes

Based on waterjet working principles, process characteristics and interactions mechanisms with tissues, medical waterjet processes can be classified as three categories: continuous, pulsed and abrasive waterjet as shown in Fig. 3.

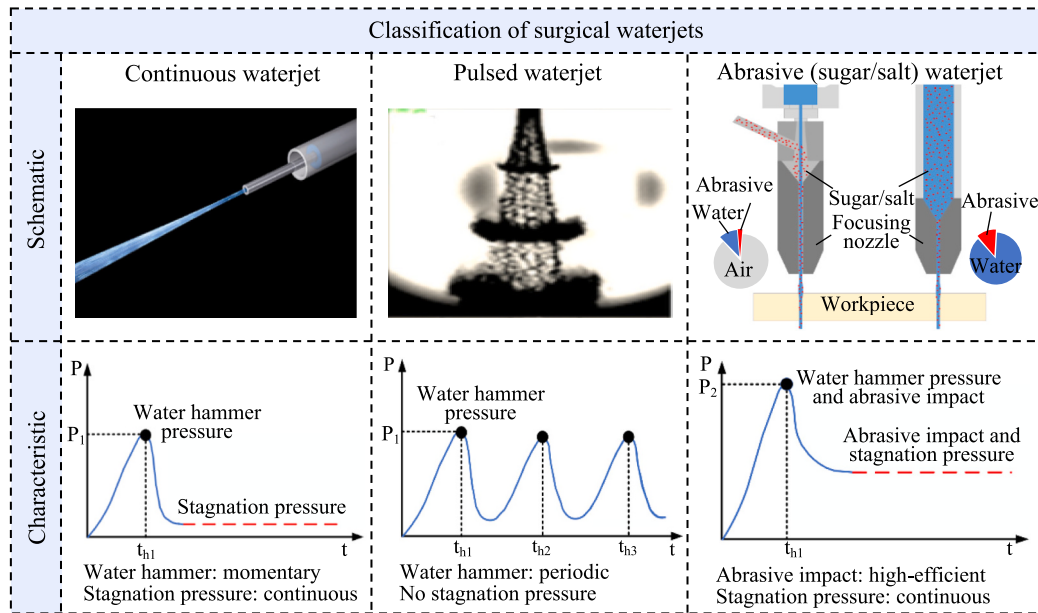


Fig. 3. Classification of surgical waterjets [31,51,52]. All figures reprinted with permissions from Elsevier.

2.2.1. Continuous waterjet

Continuous waterjet is most widely used in surgeries, particularly in separation of soft tissues because it is relatively simple and feasible to implement. It uses an uninterrupted fluid for operations at various water pressure levels, which includes the initial momentary water hammer pressure and the constant stagnation pressure with the unchanged total energy over time [53]. Continuous waterjet has achieved tissue-selective separations of normal and diseased tissues through the impact kinetic energy of stable high-speed fluids [1], which is different from the rigid cutting caused by a scalpel. The saline solution in continuous waterjet is commonly recommended as a separation medium in surgeries [54] to expand the gap of tissue structure. This is because the osmolarity of the stroke-physiological saline solution is similar to human blood, and its sodium content is similar to plasma to supply electrolytes and maintain the tension of body fluids.

Softer tissues, such as liver parenchyma, can be dissected at lower water pressures and preserve the blood vessels and bile ducts [11]. By adjusting waterjet pressures, continuous waterjet is suitable for selective separations of soft tissues with different microstructures and mechanical properties. Current continuous waterjet is usually generated by a sterile disposable double plunger pump [54], which is easier to execute than pulsed waterjet and abrasive waterjet. The process has been widely applied in surgeries, such as dissection of liver parenchyma [55], cholecystectomy [11], dissection of tumors [12], total mesolectal excision [56], and ablation of benign prostatic hyperplasia [13]. Further, continuous waterjet is also commonly used for debridement [19], in which high-velocity fluids separate necrotic tissues close to wounds.

However, there are technical issues to be addressed in continuous waterjet for surgeries. First, as a high volume of water supply is required for dissection, the consequent water vortex tends to disperse tissue debris [57]. Secondly, in continuous waterjet assisted with a neuro endoscope, there were risks in elevating of intracranial pressures and obscuring of the neuro endoscopic view during ventricle wall dissection [57]. Thirdly, continuous flow of fluid in waterjet may jeopardize precise control of cutting depths in surgeries [58].

2.2.2. Pulsed waterjet

Following the usages of continuous waterjet in surgeries, pulsed waterjet has also been introduced for soft tissue dissection and bone cutting.

Pulsed waterjet uses a discontinuous and periodic fluid as a water hammer to periodically impact tissues at frequently changed water pressures and powers [59]. In the process, the pressures have water hammer and stagnation stages [60]. In a water hammering stage, compressed water forms a strong erosion impact to target tissues at the peak pressure (i.e., water hammer pressure) to generate waterjet compression zones of high-speed collisions between the waterjet and the target tissues [60]. Compared with continuous waterjet, pulsed waterjet with the water hammer pressure enhances the striking ability to significantly improve the separation efficiency of tissues. Therefore, it achieves tissue separations at lower water pressures and less water flows to avoid serious complications often caused by continuous waterjet, such as elevations of intracranial pressures and local distortions of hypothalami in neurosurgeries [57].

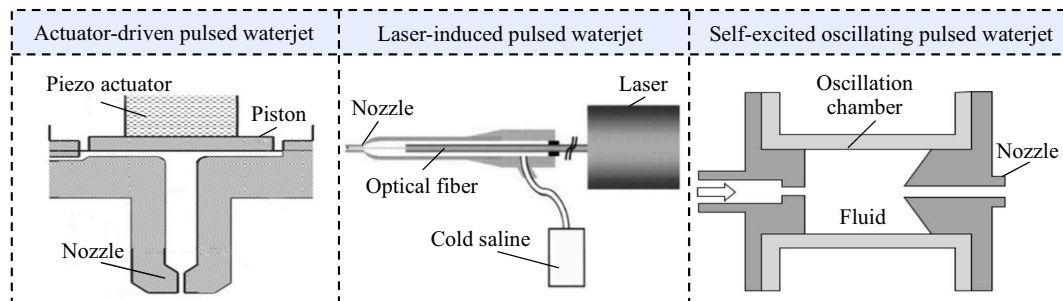


Fig. 4. Configuration principles of three types of pulsed waterjet [24,63,64]. All figures reprinted with permissions from Elsevier, Springer and AIP.

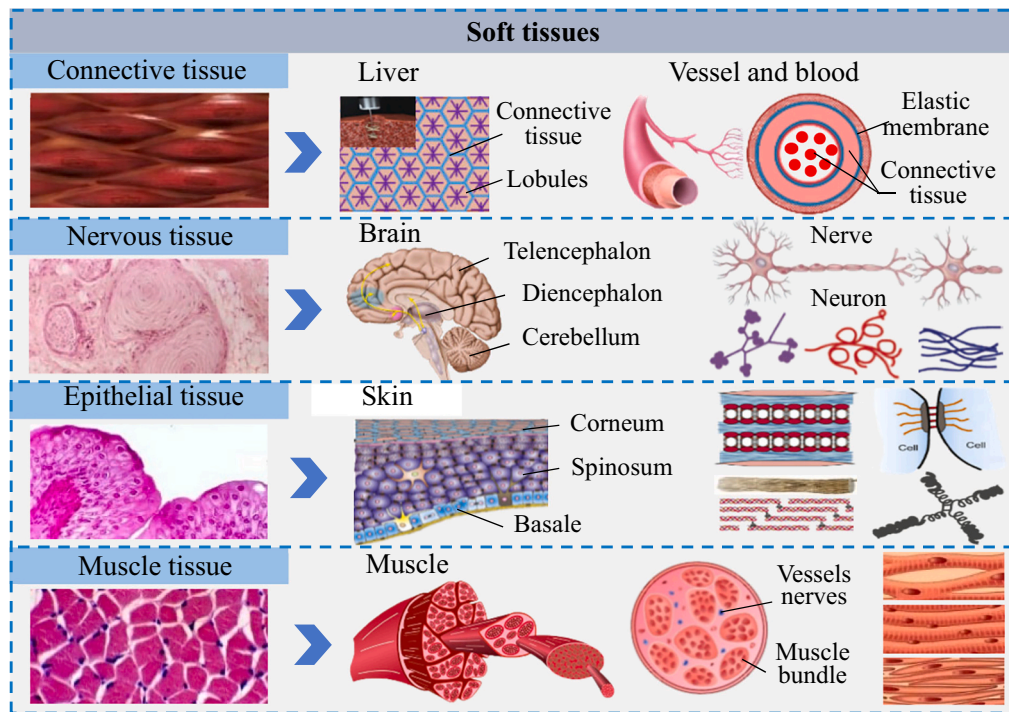


Fig. 5. Classification of soft tissues [71,73–76]. All figures reprinted with permissions from Elsevier, Springer and Wiley.

Continuous water usage may cause a vortex or fountain effect, which can lead to tissue catapulting. In particular, this leads to disseminations of pathological tissues [43] and obscuring of endoscopic views [57]. However, as no disturbance occurred in surgical fields, even in deep narrow speculums by splashing, misting, or aerosol [43], pulsed waterjet achieves finely controlled tissue dissections under clear neuro-endoscopic views, and preservations of tiny vessels of 100–200 μm for less blood loss [43]. Therefore, pulsed waterjet is more suitable for high-risk and precise brain surgeries than continuous waterjet. In addition, continuous waterjet is commonly used for submucosal injections to elevate mucosae in gastrointestinal surgeries, pulsed waterjet achieves tissue dissections with less water due to its superior processing efficiency [61]. Pulsed waterjet is experimentally used in bone cutting, which achieves higher removal rates than continuous waterjet [62], signaling its potential as a new tool for clinical bone cutting in the future.

In spite of its advantages, pulsed waterjet has technical limitations, particularly its complex configurations, including actual-driven [24], laser-induced [63] and self-excited oscillating types [64], as shown in Fig. 4. Actuator-driven and laser-induced pulsed waterjets are commonly chosen for medical usages while self-excited oscillating pulsed waterjet is more used for industry. Slight delays of fluid projections after switching on pulsed waterjets also cause minor inconvenience [63]. Further, there lack long-term safety evaluation and verification of pulsed waterjet devices for their applications in surgeries and more research efforts are required to address the issues. Additionally, shock waves in pulsed laser-induced waterjet may lead to micro-vascular rupture, negatively affecting neuro endoscopy [65], although they can transiently change the membrane permeability, benefiting drug delivery [66]. Therefore, the usage of shock waves needs to be optimized to avoid tissue damage in surgeries.

2.2.3. Abrasive waterjet

Continuous and pulsed waterjets yield low efficiencies in cutting of bone due to its high strength and hardness. In comparison, abrasive waterjet achieves higher efficiencies of abrasive erosion of the hard tissue.

Abrasive waterjet utilizes an uninterrupted mixed fluid medium

comprising sugar or salt abrasives and water for material removal to achieve high removal rates and good surface quality because of the leading role (90 %) of the abrasives with high kinetic energy in machining. As shown in Fig. 3, abrasive waterjet includes abrasive water injection jet, i.e., three phases of air, abrasive and water, and abrasive suspension jet, i.e., two phases of abrasive and water [52]. An injection jet injects the abrasives into a high-speed waterjet through a mixing chamber [52], during which approximately 20 % of the kinetic waterjet energy is directly transferred to the abrasives, causing a relatively lower energy utilization and processing efficiency. In contrast, a suspended jet pushes premixed water and abrasives in its nozzle at lower water pressures, which is often seriously worn.

In bone cutting using abrasive waterjet, the solubility of bio-abrasive needs to be considered. Due to its superior to continuous and pulsed waterjets in terms of removal rates and surface quality, abrasive waterjet may be potentially useful for prosthetic repair surgeries or osteotomies in the future. However, the selection of biocompatible and biodegradable abrasives used in abrasive waterjet for bone cutting is challenging. Several abrasives, such as sucrose [67], sodium chloride, bone powder [68], and lactose [69], have experimentally achieved higher machining efficiencies in bone cutting. However, the clinical performance of these abrasives still needs to be further studied.

3. Waterjet dissection of soft tissues

As mentioned earlier, dissection and ablation of soft tissues, such as livers, galls, bladders, brains, kidneys, and prostates, has been performed using waterjet because of its superior to traditional devices (e.g., scalpels, saws, and medical electric drills). Revolutionary waterjets have been developed for various surgeries [70]. The soft tissue properties, dissection/ablation mechanisms, processing properties, waterjet tool design, and their clinical performance are summarized as follows.

3.1. Soft tissue properties

Soft tissues are generally classified as connective, nervous, epithelial, and muscle tissues which are not hardened or calcified as shown in

Table 2
Elastic modulus and tensile strength of soft tissues.

Type of soft tissues	Elastic modulus (MPa)	Tensile strength (MPa)
Liver	0.0072–0.0121 [77]	0.0095–0.0096 [77]
Blood vessel	2.51 [78]	0.83–2.61 [79]
Brain	0.00224–0.00336 [80]	–
Skin	48.4–118.2 [81]	13.2–30.0 [81]
Muscle	0.94–2.22 [82]	0.15–0.37 [82]
Cholecyst	0.255–0.641 [83]	0.348–1.240 [83]
Colon	0.55–2.97 [84]	0.9 [85]
Kidney	0.096–0.180 [86]	0.024–0.031 [86]
Bladder	1.9–2.3 [87]	0.8–1.2 [87]
Uterus	0.005 [88]	0.255–1.140 [89]

Fig. 5. Objects in surgeries are organs which are composed of various tissues, such as liver, blood vessels, brains, muscles, etc. [71]. In the point of view of material composition, soft biological tissue is composite material, which composed of cells and extracellular matrix. Extracellular matrix is a complex network composed of a variety of macromolecules around cells, including collagen fibers, elastic fibers, and reticular fibers [71]. The content of these three main tissue components and the three-dimensional structure formed by interlacing of different biological soft tissues vary significantly [72]. Therefore, the mechanical properties and cutting characteristics of different objects in surgeries are notably different. The elastic modulus and tensile strength of some soft tissues are summarized in Table 2.

As a material with complex mechanical properties, some scholars have carried out relevant experimental and modeling work pertaining to the composite structure seen in soft tissues, as shown in Fig. 6, including skin, blood vessel, muscle, liver etc. Skin, the largest organ, is the outermost layer of a human body [90], which consist of abundant collagen fiber, elastic fiber and reticular fiber [91]. The skin is composed of epidermis, dermis and hypodermis, each of which has different mechanical properties [76]. Hendriks et al. developed a 2D bi-layered model to characterize the mechanical properties of the epidermis and

the dermis [92]. Further Tran et al. [93] made a two-dimensional model composed of three layers of skin structure and muscle using magnetic resonance imaging and used finite element method to simulate the indentation process. Groves et al. [94] embedded three groups of fiber families into an isotropic material matrix to establish a nonlinear and anisotropic model of skin by tensile tests. Blood vessel is essential to the blood circulation system, which is composed of intima, media and adventitia. It is composed of abundant collagen fiber, elastic fiber, reticular fiber and smooth muscle [95]. Holzapfel et al. [96] developed a constitutive law for the description of the mechanical response of arterial tissue, which consisted of two different layers of fiber reinforced materials. Later, Gasser et al. [97] presented an improved model using a distributed fiber orientation for each family of fibers. Lu et al. [95] developed a soft composite anisotropic constitutive model driven by fiber microstructure to describe arterial tissue. Muscle tissues includes skeletal, cardiac, and smooth muscles [73], of which skeletal muscle is the largest proportion of the body's tissue. Skeletal muscle is a composite tissue composed typically of ~ 90 % contractile muscle fibers and ~ 10 % connective tissue supporting structures [98]. Fernandez et al. [99] characterized continuum muscle mechanics though diffusion tensor imaging fibers. Gfrerer et al. [100] proposed a composite structural muscle model that includes sliding contacts between individual fibers and between fibers and bones. The liver is the second largest organ, which is a multi-unit reinforced structure with densely distributed vascular network and connective tissue [71]. Peterlik et al. [101] proposed a real-time model of vascularized liver. The model incorporates distinct constitutive laws for the parenchyma (tissue) and vessels, and establishes a coupling mechanism between these two entities.

It should be noted that tissues are living biological materials and their mechanical properties largely vary with different structures [87] and compositions [71]. Distributions of elastic fibers and connective tissue in different soft tissues significantly affect crack propagations and removal mechanisms in cutting, of course, affecting tissue-selective separations using waterjet. Further, most research has only focused on

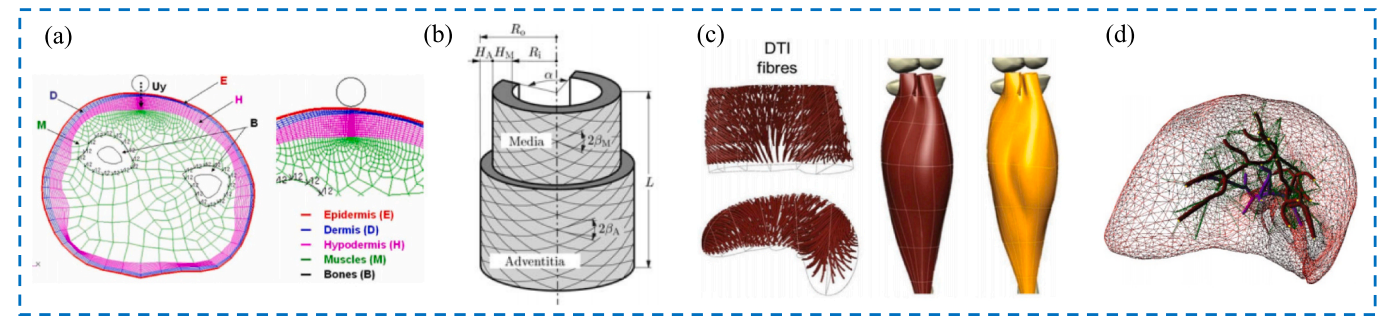


Fig. 6. (a) Two-dimensional model composed of three layers of skin [93]. (b) Arterial tissue model composed of two layers of fiber reinforced material [96]. (c) Fiber-based modeling and simulation of skeletal muscles [99]. (d) Model of vascularized liver [101]. All figures reprinted with permissions from Taylor & Francis and Springer.

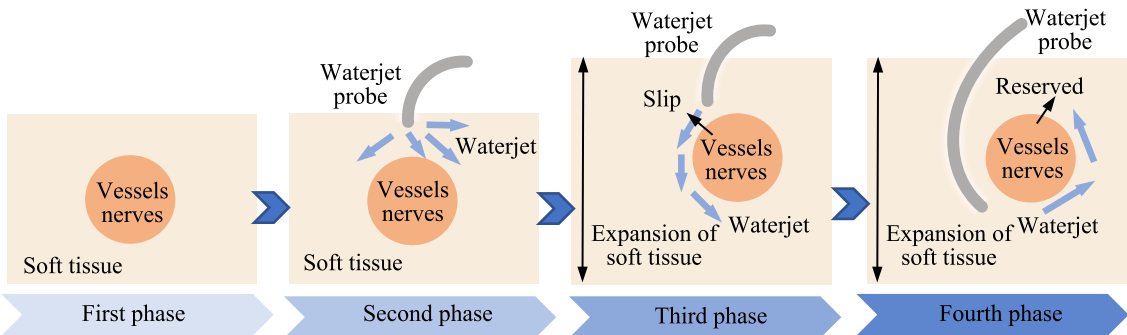


Fig. 7. Mechanisms of tissue-selective separation in waterjet cutting.

Table 3

Waterjet pressures for dissecting different tissues in surgeries.

Type of soft tissues	Dissection pressure (MPa)	Refs.
Liver	3.0–4.0	[55]
Cholecyst	3.0	[11]
Kidney	1.6–2.2	[103]
Parotid gland	3.0–5.0	[28]
Lymph node	2.0	[12]
Hystera	3.5–6.0	[22]
Rectum	5.0	[56]
Brain tumor	0.3–4.5	[104]

limited tissue types, e.g., liver, skin, and muscles. The mechanical properties of multiple tissues and their constitutive and physical models are not yet thoroughly investigated. The applicable scope and conditions of the existing composite structure models are single and the deviation of models is large, resulting in the lack of completeness and universality, which makes breakthroughs in soft tissue dissections more challenging.

3.2. Waterjet dissection mechanisms for soft tissues

The removal mechanisms and damages induced in soft tissue dissection by waterjet are determined by the inherent tissue properties and processing characteristics. The understanding of these mechanisms provides the scientific guidance for designing new waterjet tools to minimize crack propagations and failures in soft tissues.

The kinetic energy of high-speed water of waterjet involves tissue separations during dissection processes [1]. In the tissue-selective separations shown in Fig. 7, the intervention gap of the waterjet medium causes the soft tissue expansion. For example, the liver parenchyma is dissected at 3.0–4.0 MPa water pressures [55]. At such pressures, stronger, tougher and more elastic tissues, such as blood vessels and nerves, slip the waterjet impact and remain intact because of their high proportions of collagens and elastic fibers in the compositions. Membranous structures are also resistant to the waterjet impact [102] with a successful example of selective dissection of brain tumors adjacent to the optic nerve using waterjet because the arachnoid membrane is much stronger than gray and white matters of the brain [23]. Table 3

summarizes the waterjet pressures for difference soft tissues applied in surgeries. This indicates the importance of the waterjet process selection for successful tissue-selective separations with maximum protections of healthy tissues.

The waterjet-tissue interactions have been studied to comprehend the dynamic evolution of soft tissue dissection [1,53], featuring a non-rigid impact to soft tissues with high-speed waterjet. As shown in Fig. 8(a), a waterjet tip contacts the soft tissue surface when commencing the water injecting [1]. Then, waterjet shock waves are formed at the liquid-tissue contact, and respectively propagate in water and soft tissues at different speeds. This forms a dense compressed water region in the waterjet, leading to the elastic deformation of the soft tissue. As the water beam continuously impacts the tissue surface, new shock waves are formed and their propagation speeds are much faster than previous waves, resulting in a water hammer pressure. The pressure increases with the continuous water injection to form a dynamic pressure, which eventually evolves into a steady Bernoulli stagnation pressure. During waterjet processing, the soft tissue undergoes the elastic deformation to fracture damage, which can be described in deformation, destruction, separation phase, and recovery phases because of its creep, relaxation, and hysteresis properties [53]. As shown in Fig. 8(b), the deformation phase specifically involves the waterjet compression-induced elastic deformation and edge distortion of the tissue. The compression expands to increase invasive forces F of the waterjet and the waterjet shock energy is absorbed by the tissue tension and expansion effect. In the destruction phase, the tissue deformation has little fluctuation at its maximum strain, and the waterjet impact energy absorbed by the tissue is released instantaneously to create cracks. The waterjet kinetic energy conversion rate determines the cracking surface area [53]. The more energy is absorbed by the tissue in the previous phase, the larger cracking surface area is generated in this phase, leading to the separation phase during which cracks extend to separate the tissue. In the recovery phase, the waterjet impact stops and the soft tissue stores the elastic potential energy. Once the energy is released, and the soft tissue regains its deformation.

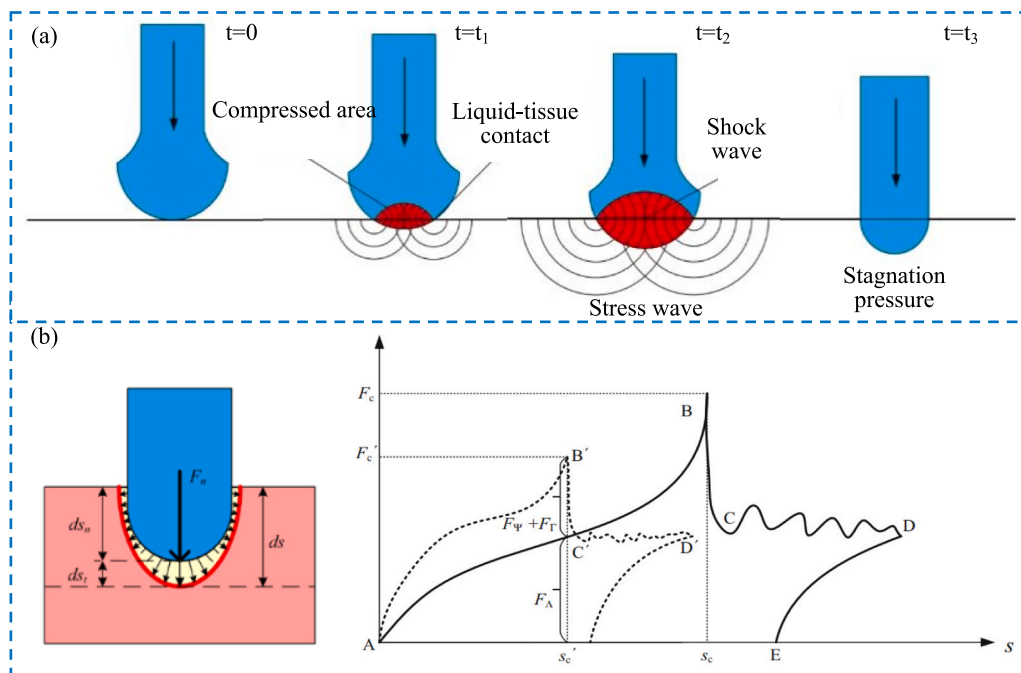


Fig. 8. (a) Soft tissue deformation and movement under waterjet hammer impacting forces [1]. (b) The force-displacement relationship of liquid-tissue contact during soft tissue separation [53]. All figures reprinted with permissions from Springer and IEEE.

Table 4
Summary of waterjet dissection mechanisms for soft tissues.

Author	Contribution	Tissue	Process parameter
Alitavoli et al. [105]	Effects of the process parameters on the cutting depth	Pork, beef, lamb, chicken	Pressure, traverse speed, nozzle diameter, stand-off distance
Baxter et al. [108]	Depth prediction model of water jet penetrating the skin	Skin, hydrogel	Fluid density, outlet velocity and nozzle diameter
Shergold et al. [109]	Shergold-Fleck sharp-tipped punch penetration model	Human skin, rubber	Waterjet diameter and waterjet stagnation pressure
Yamada et al. [102]	Selective dissection of soft tissue by pulsed waterjet	Swine liver	Input voltage, moving velocity and breaking strength of liver
Tagawa et al. [110]	Viscosity stress and repulsive force model	Gelatin, human skin	Waterjet velocity, penetrate time
Shi et al. [106]	Effects of the process parameters on the cutting performance	Porcine and rat liver	Pressure, impact angle, nozzle diameter, stand-off distance
Morad et al. [111]	Waterjet cutting depth prediction model	Beef	Pressure, traverse speed, nozzle diameter, stand-off distance
Seok et al. [7]	Effects of process parameters on waterjet penetrating the cadavers	Human cadavers	Pressure, Young's modulus
Bahls et al. [107]	The linear relationship between pressure and cutting depth	Gelatin	Pressure, material properties
Babaiasl et al. [25,49]	FEM and physical prediction model of waterjet cutting depth	SEBS	Tissue stiffness, waterjet velocity and nozzle diameter
Cao et al. [112]	SPH-FEM model of waterjet cutting soft tissue	Porcine liver, gelatin	Pressure, impact angle, traverse speed

3.3. Process parameter effects on dissection performance

Based on the soft tissue dissection mechanisms, the effects of waterjet processes on the dissection have been investigated through both experiment and simulation approaches.

Table 4 summarizes experimental studies of waterjet processes for soft tissue dissection. In high-pressure waterjet with sugar and salt abrasives for meat cutting (e.g., pork, beef, etc.), more abrasives improve the cutting efficiencies [105]. Specifically, higher pressures and larger nozzle diameters tend to result in greater cutting depths. While increased stand-off distances and traverse speeds typically lead to decreased cutting depths [105]. In tissue-selective separation for pig liver dissection using a piezo actuator-driven pulsed waterjet [102], the cutting depths increase with the input voltage, and decrease with the cutting speed and the lever strength [102]. As the liver parenchyma strength is significantly lower than those of hepatic veins and Glisson's sheaths, the histological staining proves that the former is selectively dissected and the stronger latter are preserved [102]. This reconfirms the controlling role of waterjet pressure in tissue-selective separation in liver dissection [106]. Meanwhile, repeated waterjet with quick traverse speeds also improve cutting depths [107] and avoid circumferential radiation damage [106] and the cutting capacity decreases with increased impact angles [106], which aggravate waterjet reflections and reduce impact forces [106].

In smooth-particle hydrodynamics finite element method (SPH-FEM) modeling of high-speed waterjet separation of soft tissue, as shown in Fig. 9(a), the strain and fracture thresholds for soft tissue decrease with increased impact pressures, producing less damage in dissection [112]. In modeling of waterjet cutting/penetration depths in soft tissues as shown in Fig. 9(b), waterjet penetrations to the tissues reveal erosion, stagnation, and dispersion mechanisms [113]. The model was established using turbulent theory to predict penetration depths of waterjet erosion in skin and polyacrylamide gel [108]. However, the prediction

error for polyacrylamide gel is 20 % while the model for skin is limited by a certain stand-off distance [108]. The Shergold-Fleck sharp-tipped punch penetration model was also established for waterjet penetration of soft tissue [109], which indicates that the penetration mechanism for soft tissues is similar to penetrating a soft solid with a sharp tip to form planar cracks. Waterjet penetration of gelatin and human skin was also simulated to construct the viscous stress model and the repulsive force model, as shown in Fig. 9c [110]. The penetration depths into gelatin experimentally and numerically increased linearly with the jet velocity. The penetration depths into skin also increased with the jet velocity. Considering skin surface resistances, the combined viscous stress and repulsive force models achieved better predictions for penetration depths [110].

The waterjet penetration depths into cadaver cheeks were also measured, showing their increase with the jet pressure and decrease with the Young's modulus of the tissue [7]. As shown in Fig. 9(d), the effects of soft tissue stiffness, water flow, and nozzle diameter on cutting depths into soft tissues were also been experimentally and numerically studies [25]. The depths were found to be proportional to the flow rate and inversely proportional to the tissue stiffness and the nozzle diameter [25]. In particular, the depth of cut showed almost a linear relation with the flow rate when the waterjet nozzle was sufficiently small. The model was verified experimentally with styrene-ethylene-butylene-styrene (SEBS) soft tissue simulants [49].

All these studies provide insights into the property-process relations in soft tissue dissection with waterjet. However, existing numerical prediction models are only valid in limited experimental conditions for certain soft tissues. In addition, the understanding of soft tissue removal mechanisms in waterjet is still in its infancy.

3.4. Waterjet dissection in open surgeries

Open surgery is the most common surgical procedure. Since the first waterjet dissection of the liver mentioned in Fig. 2, the techniques, as summarized in Fig. 10, have been used in many open surgeries for organ transplantations, e.g., liver transplantation [115], brain tumor dissection [104], intraocular surgery [32], and resection of sweat glands [10].

3.4.1. Waterjet dissection in hepatectomy

Traditional liver surgeries for resections and transplantations often cause intraoperative bleeding. To preserve blood vessels in liver surgeries, waterjet dissection developed in 1982 [3], has been evolved from case studies and small series initiated in Europe [116], to a large number of clinical surgeries reported by Vollmer et al. [115]. In hepatectomy, a fine, high-pressure water flow enables the selective separation of surrounding connective tissue components from blood vessels. The water displaces the fibers and collagen tissue, leaving these structures intact. Therefore, waterjet can be utilized for treating the cirrhosis, hepatocellular carcinoma, and solid hepatic tumors [71]. The techniques are more precise, controllable, flexible tissue-selective and less aggressive to blood vessels. For instance, a waterjet with a nozzle diameter of 0.08–0.20 mm and operating at a pressure of 50 bar has the potential to preserve blood vessels as small as 0.2 mm during a hepatectomy [117]. Compared with other tools (e.g., stapler, CUSA, blunt dissection), waterjet cutting during hepatectomy operations has shown an increase in rate of complete tumor resection by 5–19 % [118], a decrease in blood loss per resection plane by 20–62 mL/cm² [55], a decrease in blood transfusion requirement by 50 % [55], a decrease in complication rates by 14 % [115], a decrease in Pringle-time per resection plane by 50 % [55], a decrease in operation time by ~ 1 h [119]. These investigations demonstrate that waterjet could reduce blood loss, complication rates and recovery time. However, technical limitations with waterjet still exist, including air bubbles created by the process, which affect view fields during surgeries, and back sprays of blood and liquid during operations, which can cause problems in the surgeries [120].

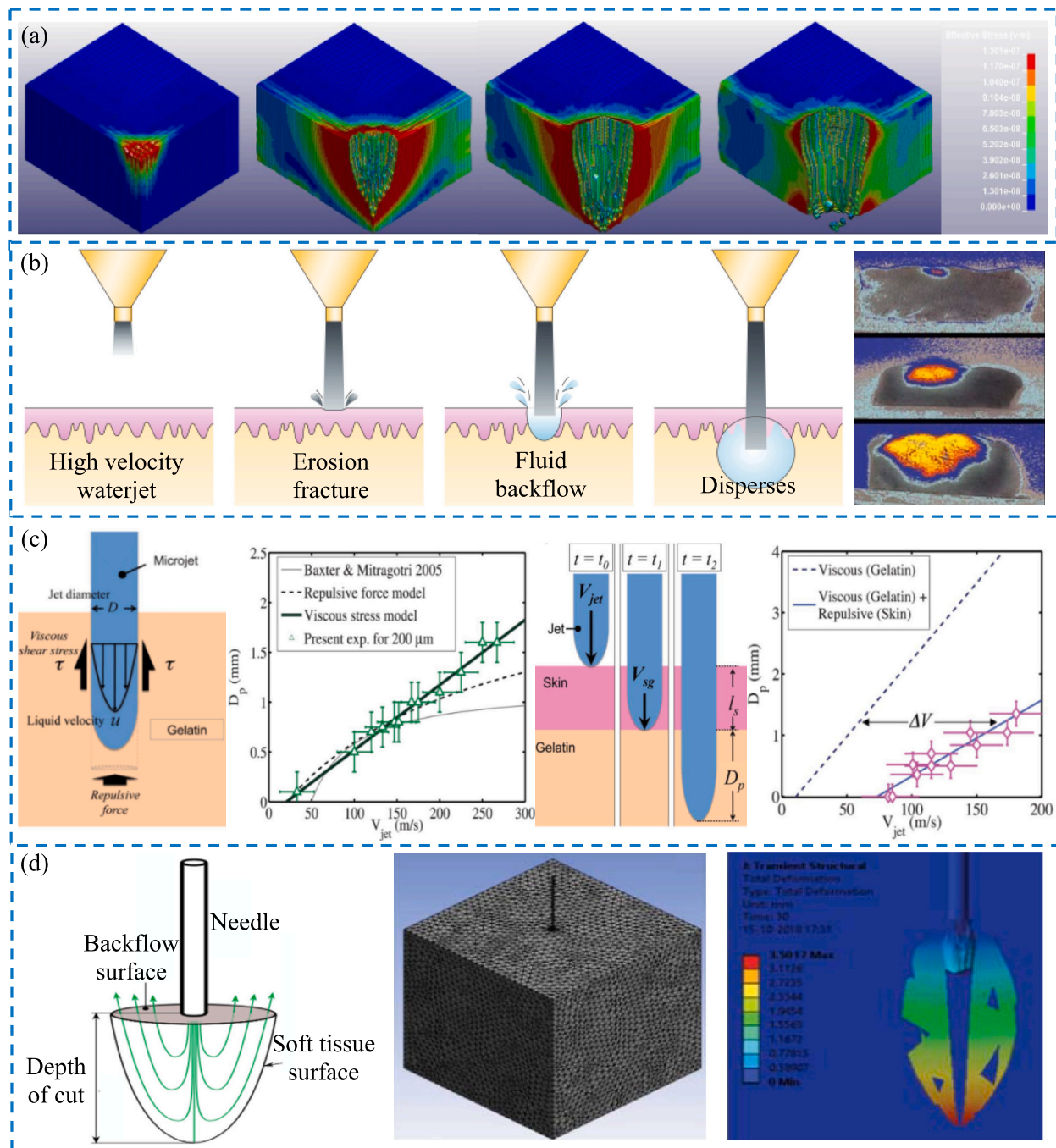


Fig. 9. (a) SPH-FEM model of waterjet dissection [112]. (b) Waterjet penetration through the skin [113,114]. (c) Model of waterjet penetrating gelatin and human skin-enclosing gelatin [110]. (d) FEM prediction of cutting depths in waterjet cutting of SEBS [25,49]. All figures reprinted with permissions from Elsevier, Springer, Royal Society of Chemistry and IEEE.

3.4.2. Waterjet dissection in neurosurgery

In brain surgeries, damage to cerebral vessels may lead to hemiparesis and aphasia in patients. In particular, thermal damage to brain tissues occurs when temperatures reach 43 °C or above for 60 min [63]. Obviously, waterjet is a superior tool in neurosurgery to avoid the thermal problem. Since waterjet was introduced for neurosurgery by Terzis et al. [35] in 1989, the evolved technique enabled the preservation of blood vessels larger than 20 μm in the porcine brain and more progress has been made by the Ernst Moritz Arndt University in Germany [27]. Waterjet can precisely separate tumors from brains, while preserving parenchymal borders of the tumors, and nerve morphologies and functions and improving radical resections of tumors.

For neurosurgery, there are low-pressure (0.003–0.01 MPa) and high-pressure (0.8–1.2 MPa) waterjets available. The former is commonly used to separate tumors, such as meningioma and aneurysms, from the cortex in a mild way [121]. The waterjet was used to gently separate the tumor from the arachnoid-covered brain surface by pulling on the free edge of the tumor [122]. The previous report has demonstrated that low-pressure water separation could serve as an effective alternative to retractors in routine microsurgery [121]. It was found to be more flexible and gentler in its approach. However, it is not powerful enough to separate firmly adherent and invasive tumors. In contrast, high-pressure (0.8–1.2 MPa) waterjet is utilized as a non-thermal knife for cutting and separating lesions from brain tissue, and is widely

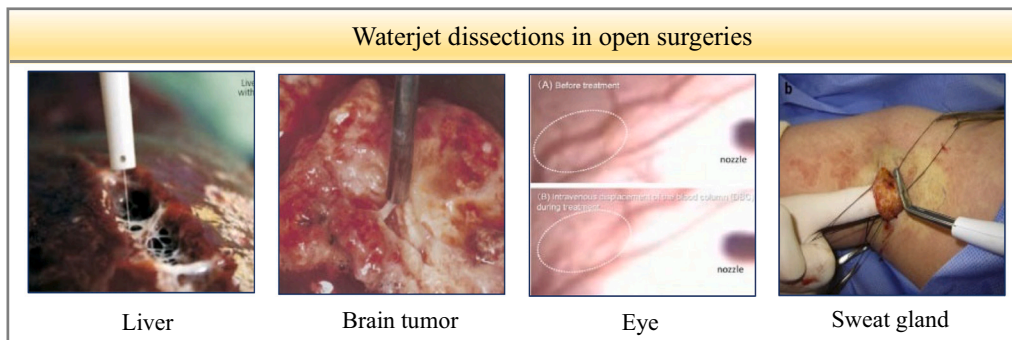


Fig. 10. Applications of waterjet dissections in open surgeries [4,10,21,32]. All figures reprinted with permissions from Elsevier, Springer and Assoc Research Vision Ophthalmology Inc.

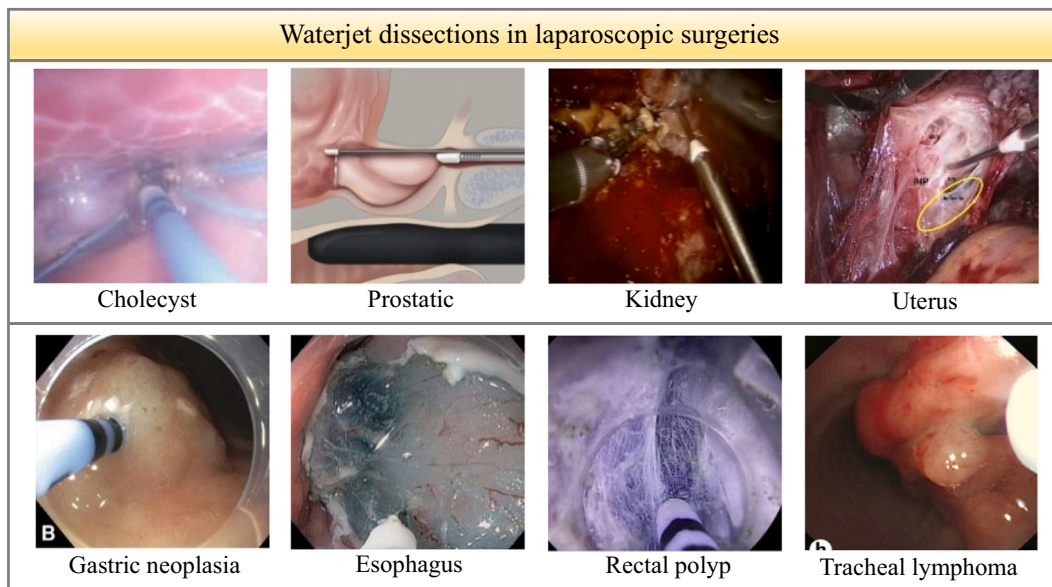


Fig. 11. Applications of waterjet dissections in laparoscopic surgeries [5,125–131]. All figures reprinted with permissions from Elsevier, Springer, Karger, Thieme and Baishideng.

applied for removal of neurosurgical lesions, such as gliomas, epilepsy, hemangioblastoma, and metastases [121]. In neurosurgery, it is often challenging to determine the exact separation plane between tumors and healthy tissue using conventional surgical tools. The inaccurate excision may leave behind residual lesions, causing harm to healthy tissue and even leading to recurrence. However, waterjet technology allows for precise tumor resection by controlling the pressure, while preserving the integrity of the brain's arteries, veins and arachnoid [123]. Waterjet is capable of excising hard and well-defined tumor lesions [104] as well as separating soft and poorly defined metastatic tumor lesions [27]. For example, the dissection of glioma requires pressure of 0.3–1.7 MPa; metastasis of 0.5–1.0 MPa; meningioma of 1.5–4.5 MPa; epilepsy of 0.4–1.0 MPa; ICA stenosis of 0.5–4.5 MPa; hemangioblastoma of 0.4–1.0 MPa [104]. Compared with ultrasonic aspiration in epilepsy surgery, waterjet has shown a decrease in blood loss by ~ 50 mL and operation time by ~ 9 min [124].

3.5. Waterjet dissection in laparoscopic surgeries

Following the positive clinical outcomes of waterjet for open surgeries, the techniques have also been used in laparoscopic surgeries as minimally invasive operations for complete resections or ablations of diseased tissues with less trauma and shorter convalescence time. As shown in Fig. 11, the combination of waterjet and laparoscopic surgery

has been widely used in many surgeries, such as dissections of various tumors [125] and ablations of benign prostatic hyperplasia [13].

3.5.1. Waterjet dissection of tumors

The advantage of laparoscopic surgery makes it more suitable for tumor dissection. Endoscopic mucosal resection and endoscopic sub-mucosal resection are now the standard procedures for minimally invasive treatment of superficial tumors [132]. However, perforation pain and bleeding are common complications during tumor dissection surgeries using conventional surgery tools. To minimize such complications, waterjet dissection and waterjet-assisted knives have been used in surgeries for liver tumors [36], cervical carcinomas [125], partial nephrectomy [131], and testis cancers [12], as shown in Fig. 10. In these operations, waterjet dissection and waterjet-assisted knives have effectively shortened the operation times and avoided endoscopic accessory-switches, reducing bleeding, perforation rates, and tumor recurrence. Consequently, the waterjet technique improves the safety and efficiency of resection compared to conventional endoscopic knifing techniques. In addition, waterjet dissection can selectively separate fibrous and connective tissue to protect blood vessels [11], fine lymphatic vessels, and nerves [12], thus preserving organ functions, such as urodynamics [125] and antegrade ejaculation [12]. All these operations have demonstrated the benefits of waterjet techniques in dissection of tumors in laparoscopic surgeries [36].

Table 5

Waterjet devices in soft tissue dissections. All figures reprinted with permissions from Elsevier, Springer, Assoc Research Vision Ophthalmol Ogy Inc. and Georg Thieme Verlag KG.

Country and manufacturer	Device	Description	Refs.
Germany-ERBE		Pros: Generalization, high precision, wide range of waterjet pressure, spiral waterjet, mature technology. Cons: Instruments need to be replaced during surgery lead to more time.	[27]
Germany-ERBE		Pros: Generalization, multi-function (combined with VIO electrosurgery station), high precision, wide range of waterjet pressure, less procedure, spiral waterjet, mature technology.	[31]
USA-PROCEPT BioRobotics		Pros: Waterjet ablation combined with ultrasound-guided and robotic-assisted, irreplaceable, high precision, highly targeted. Cons: Poor versatility.	[13]
Japan-Tohoku University		Pros: Pulsed waterjet, better cutting performance, less water consumption, small size. Cons: Limited adjustment range and lower stability, pressure and pulse frequency change simultaneously.	[32]
France-NESTIS		Pros: Both pressure and flow can be adjusted, simple structure. Cons: Less pressure adjustment range, poor versatility.	[44]
China-AMMS		Pros: Small size, less weight, high pressure to cut soft tissue and low pressure to debridement. Cons: High noise, low efficiency, low precision, high pressure fluctuation.	[136]

3.5.2. Waterjet ablation of benign prostatic hyperplasia

Benign prostatic hyperplasia (BPH) is a common disease for aged males, and often results in bothersome lower urinary tract symptoms [133]. Although it is typically not life-threatening, it can significantly impact the quality of life [134]. There are numerous options for surgical management of BPH associated with prostates of small to moderate volume (<80 cc), which successful rates vary [134]. Over the last decade, the combination of waterjet ablation, ultrasound guidance, and robotic assistance has become a new method for the treatment of benign prostatic hyperplasia. Desai et al. [133] reported the successful application of the first-generation system, called Aquabeam, for prostatectomy in dog models in 2015. After that, many researchers reported clinical outcomes of waterjet ablation of enlarged prostates. Compared with other techniques, such as holmium laser enucleation, transurethral resection, and photoselective vaporization, waterjet ablation can be operated more quickly for both small-to-moderate and large-to-very-large prostates [135]. The technique effectively results in rapid recovery and better retention of urodynamic and sexual functions of prostates, regardless of their sizes [135].

3.6. Waterjet devices in soft tissue dissection

Medical waterjet typically utilizes a high-pressure, stroke-physiological saline solution produced by a high-pressure pump. This solution travels through a hose to an operating handpiece, which then ejects the solution through a tiny nozzle. There are several commercially available waterjet devices, such as Helix Hydro-Jet [27] and Hybrid-Knife® [31] from ERBE, a global leader in waterjet devices for soft tissue dissection of liver, brain, cholecyst, kidney, etc. As mentioned previously, Aquabeam [13] developed by Procept BioRobotics, USA, guided by ultrasound imaging, is commonly used to treat benign prostatic hyperplasia. Piezoelectric actuator-driven pulsed waterjets have also been developed for retinal vascular surgery by Kunikata et al. of Tohoku University, Japan [32]. Other developed devices include France-made NESTIS SAS [44], and China-made ADSP-BF561 [136], as summarized in Table 5 on various waterjet dissection devices.

Compared with conventional tools (e.g., scalpel, ultrasonic suction, and electrodesiccation), the advantages and disadvantages of waterjet are listed in Table 6 with respect to blood loss, operation time, tissue selectivity, thermal damage, perforation rate, and complication/sequelae for different organs. It is well known that high surgical efficiency, less invasive trauma and fast recovery are clinical goals in medicine. For instance, less blood loss and fewer complications (e.g., liver abscess) are desired in waterjet cutting of liver tissue. Precise identification of nerves and complete separation of focal boundaries are critical surgical objectives for waterjet dissection of brain [27,104]. Preservation of organ function (e.g., sexual function) is desired in

Table 6

Comparison between waterjet and traditional devices in soft tissue dissections.

Tissue type	Advantage of waterjet					
	Blood loss	Operation time	Tissue selectivity	Thermal damage	Perforation rate	Complication /Sequelae
Liver	Less [3,55,115]	Less [55,116]	More [3,116]	–	–	Less [55,115]
Cholecyst	Less [11]	Less [11]	More [11]	–	Less [11]	Less [11]
Brain	Less [9,104]	–	More [27,104]	Less [54]	–	Less [27,104]
Prostate	Less [134]	Less [134,135]	More [135]	Less [13,135]	–	Less [13]
Kidney	Less [103]	Less [138]	More [103,137,138]	Less [103,137]	–	Less [138]
Uterus	Similar [125]	–	More [22,125]	–	–	Less [22,125]
Rectum	Similar [20]	Less [20]	–	–	Less [20]	Less [20]
Gastric	Less [132]	Less [20,132]	–	–	Less [20]	Less [20]
Esophagus	–	Less [20]	–	–	Less [20]	Less [20]
Testis	–	–	More [12]	–	–	–

Limitation of waterjet

1. Blurred vision and cross infection due to back spray of blood and liquid.
2. Compromised transmission stability and control accuracy for waterjet pressure in medium supply systems.
3. High costs for disposing waterjet waste.

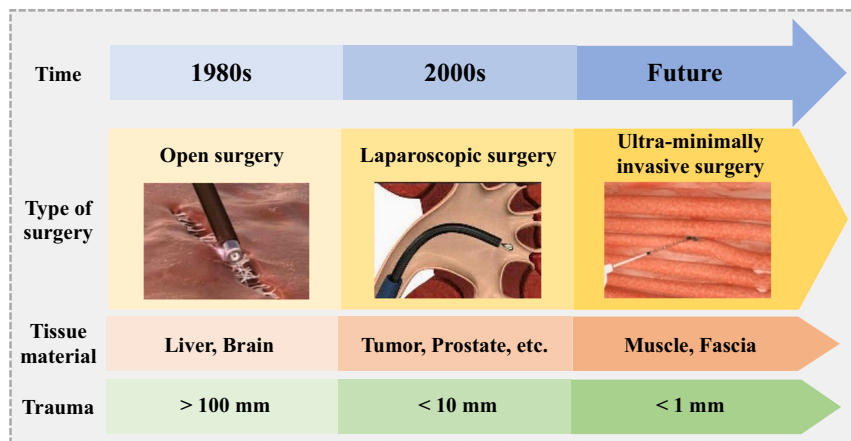


Fig. 12. Development trends of waterjet in surgeries.

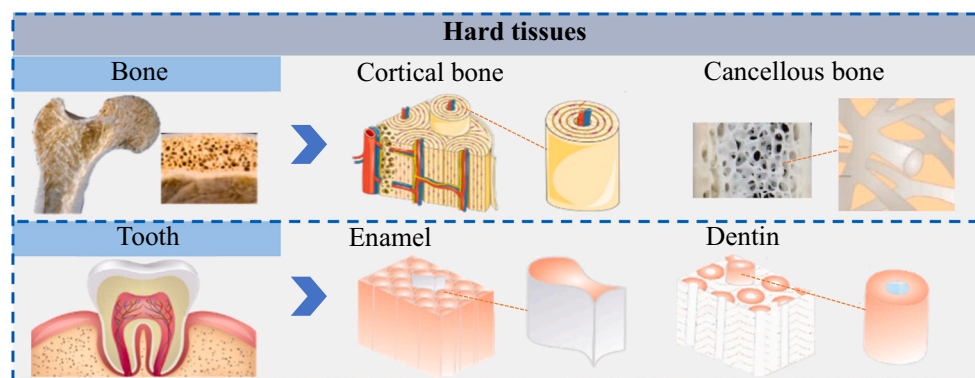


Fig. 13. Classification of hard tissues [2,139,144]. All figures reprinted with permissions from Elsevier.

prostate dissection surgery [134]. Compared with traditional devices, waterjet dissection achieves excellent identification of intrahepatic vascular structures, particularly the main branches and segmental tributaries of the portal and hepatic venous system [115]. Tissue selectivity of waterjet ensures the maximum protection of nerve tissue in the brain and prostate to reduce sequelae [104,135]. It was reported that lower perforation rate and higher resection efficiency were obtained in waterjet dissection of gastrointestinal tumors than in common electro-surgical knife (e.g., triangle tip knife, dual knife, insulated tip knife) dissection [20]. It should also be noted that thermal damage is expected to avoid in all surgeries. The investigations on renal surgery and ablation of the prostate have shown waterjet dissection of soft tissue induced the least thermal damage, enables tissue-selective separation and prevents blood loss. It was reported that renal dysfunction and broad necrosis of tissue due to thermal damage were avoided in waterjet dissection of kidney compared with radio knife and soft coagulation [137]. And lower rates of adverse events (e.g., sexual/ejaculatory dysfunction, urinary incontinence) were obtained in waterjet ablation of benign prostatic hyperplasia than transurethral resection of the prostate [13]. In addition, the technical advantage of waterjet may achieve the dissection of myofascial adhesion with needle-level trauma, significantly reducing surgical injuries from centimeter to millimeter scales. From open surgery to minimally invasive laparoscopic surgery, waterjet is being advanced for use in ultra-minimally invasive surgeries in the near future, as shown in Fig. 12.

4. Waterjet cutting of hard bone tissues

Waterjet technology has also gained increasing interest in hard tissue processing following its application in soft tissue dissection, primarily

due to its superior advantage of minimal thermal damage in comparison to conventional cutting tools. This characteristic has propelled its adoption and exploration as a viable technique for cutting hard tissues.

4.1. Hard tissue properties

Hard tissues include bone and teeth (Fig. 13), are calcified inorganic and organic matters, and water [2]. Bone is an anisotropic bio-composite composed of minerals and collagen fibers and has high strength, hard and brittle behaviors. It provides structural support for a human body for activities and protects internal organs and cells. It exhibits complex structural characteristics and diverse mechanical, biological, and chemical properties at multiple scales. Bone has cortical and cancellous structures. Cortical bone, which forms a dense outer layer of bones [139], has low porosity and high strength, and a quasi-brittle character. Cancellous bone, which forms an inner layer of bones, carries blood vessels and nerves [2] and is sensitive to thermal damage, has low strength and high porosity. The elastic moduli of cortical and cancellous bones are approximately 16.7 GPa and 1 GPa, respectively [140]. Teeth include enamels and dentins. Enamels, which are the hardest and stiffest wear-resistant outermost layers of tooth crowns, are the most highly calcified tissues with in human bodies. Dentins occupying major parts of teeth, are the elastic foundations for enamels [141]. The elastic moduli of enamels and dentins are approximately 70 GPa [142] and 17 GPa [143], respectively.

Thermal damage to bone is a critical and unresolved challenge in hard tissue processing. Waterjet machining induces least thermal damage to protect biological properties of bone [69]. Therefore, current research on waterjet machining of hard tissue all focuses on bone cutting.

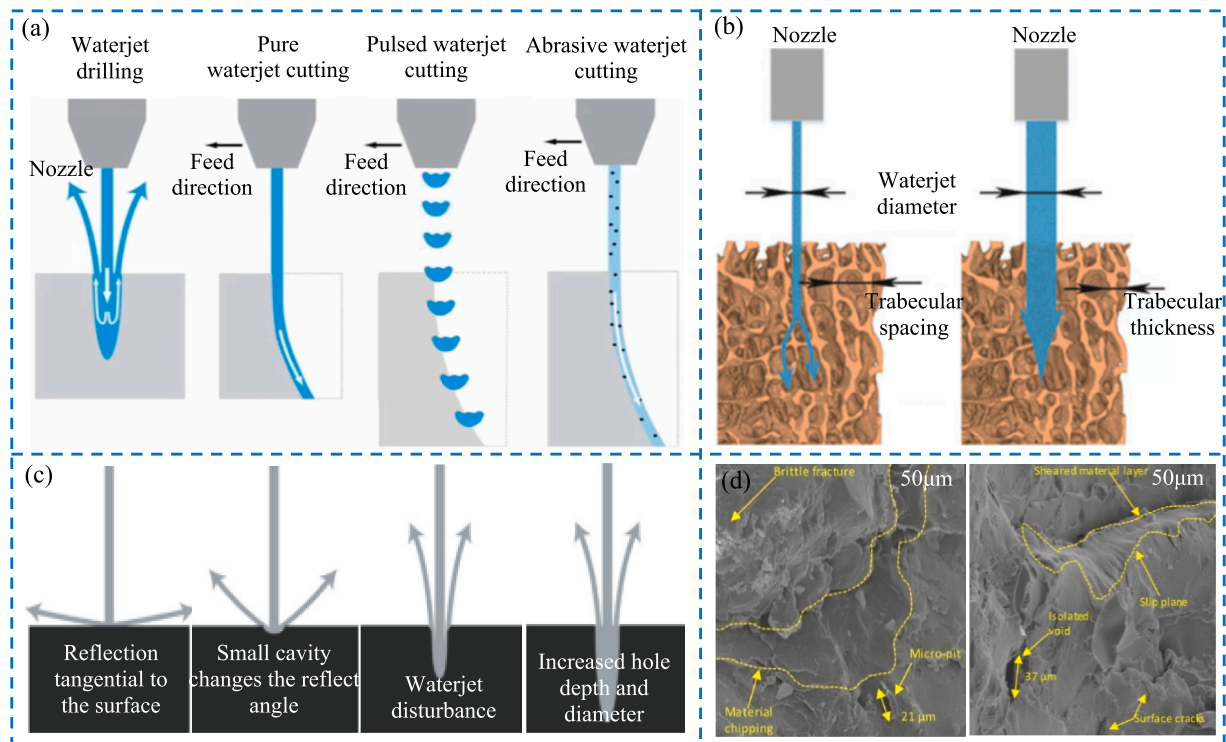


Fig. 14. (a) Characteristics of waterjet drilling and cutting of hard tissues [146]. (b) Waterjet cutting of cancellous bone [147]. (c) Different stages in waterjet drilling of articular bone [145]. (d) SEM micrographs of pulsed waterjet-cut bone surfaces [59]. All figures reprinted with permissions from Elsevier, Springer, Copernicus and AMES.

4.2. Waterjet cutting mechanisms for bone

The complex structures and properties of bone play a crucial role in determining the intricate material removal and damage mechanisms

involved in waterjet cutting. The process utilizes a high-speed medium, typically saline solution [14] or water with added abrasives [68], to generate high kinetic energy for impacting and removing the material [6]. The cutting edge of the waterjet remains sharp and clean

Table 7
Waterjet processing characteristics in bone cutting.

Authors	Waterjet	Medium	Bone	Pressure (MPa)	Nozzle diameter (mm)	Processing characteristics
Shakouri et al. [68]	Continuous	Water	Bovine femur	350	0.35	Surface quality: Ra: 15.93–16.49 μm Rz: 74–75.45 μm
Shakouri et al. [68]	Continuous	Water+Sugar	Bovine femur	150–350	0.9	Surface quality: Ra: 3.87–8.56 μm Rz: 19.72–48.57 μm
Shakouri et al. [68]	Continuous	Water+Bone powder	Bovine femur	150–350	0.9	Surface quality: Ra: 7.36–14.67 μm Rz: 54.76–80.92 μm
Biskup et al. [67]	Continuous	Water+Garnet	Bovine tibias	350	0.25	Temperature: 37–55 °C Depth: 23–64 mm
Biskup et al. [67]	Continuous	Water+Magnesium	Bovine tibias	350	0.25	Temperature: 37–60 °C Depth: 23–62.5 mm
Biskup et al. [67]	Continuous	Water+Sucrose	Bovine tibias	350	0.25	Temperature: 62–68 °C Depth: 25–67 mm
den Dunnen et al. [145]	Continuous	Water	Bone	20–120	0.6	Depth: 0–27 mm
Kok et al. [14]	Continuous	Saline solution	Goat bone	50	0.4	4–6 holes ICRS: 9–9.5
Honl et al. [30]	Continuous	Water+ Disaccharide	Bone	20–60	0.3	Depth: 0.8–2 mm
Schwieger et al. [69]	Continuous	Water+ α-lactose	Porcine femoral condyles	35–70	0.2	Surface quality: Ra: 0.03–0.14 mm
Bach et al. [15]	Continuous	Water+ Barton	Cortical bone	20–60	0.3	Depth: 3–15 mm Tolerance: 0.15–0.33 mm
Hloch et al. [149]	Pulsed	Water	Bone cement	8–20	0.7	Depth: 0.5–2.8 mm
Hloch et al. [150]	Pulsed	Saline solution	Bone cement	7–10	0.3	Depth: 0.91–1.31 mm

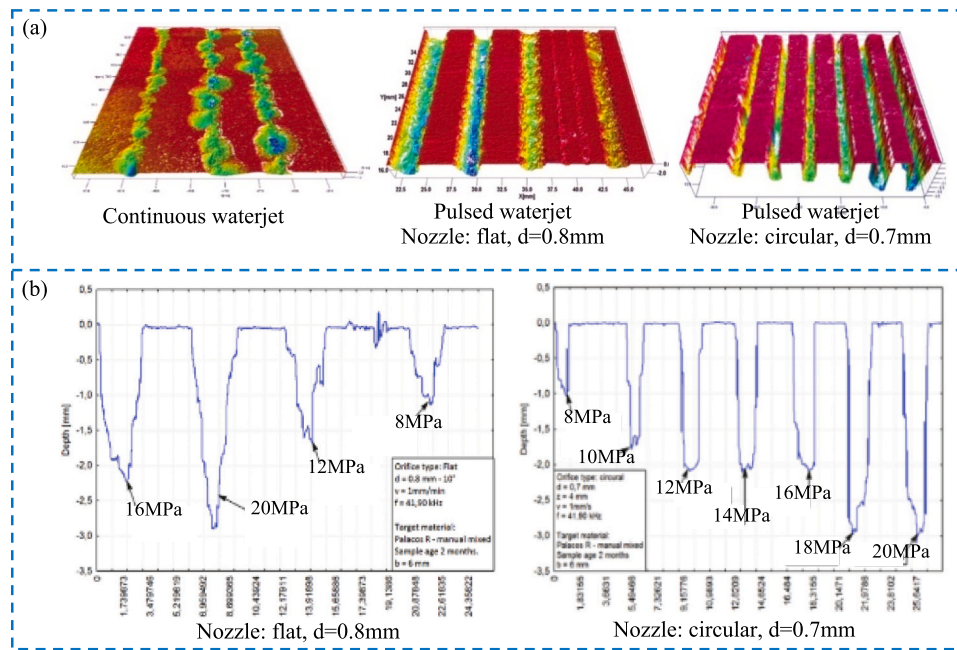


Fig. 15. (a) 3D topographies of waterjet-cut bone cement surfaces [62]. (b) 2D depth profiles of waterjet-cut bone cement surfaces with different nozzles [62]. All figures reprinted with permissions from UNIV OSIJJEK.

throughout the machining process [145]. Waterjet-induced crack initiation, propagation and failures in bone are key fundamentals for improved tool design and processing.

Fig. 14(a) shows waterjet drilling in which the incoming and outgoing flow interfere causes the energy of incoming jet to be impaired to make a shallow hole in bone. Waterjet passes through bone without the interference with back-splashing water to generate a deeper cut during the cutting process compared to drilling [146]. Fig. 14(b) depicts waterjet drilling of cancellous bone. Initially, the medium spreads tangentially into the bone due to insufficient waterjet energy and removes a small amount of bone. Continuing drilling makes a cavity and changes the reflection angle of the waterjet medium [145]. Incoming jet enters the medium-filled cavity to generate water flow disturbances and reduced flow velocities. The waterjet energy dissipates by pushing the excess medium towards the circumference and the exit of the cavity, thereby widening the cavity. Increasing the cavity depths, the waterjet energy partially dissipates, causing the excess medium to be pushed out at lower velocities. As a result, the bottom cavity diameters increases more slowly than the top ones to generate a conical shape of the cavity. This is potentially valuable for orthopedic operations, such as screw fixation or bone marrow stimulation [145]. Fig. 14(c) shows different waterjet stages in drilling of articular bone in which the medium is prone to diverge, reducing cutting efficiency, when the small diameter waterjet is used for cutting cancellous bone [147]. In general, the pressure required for cutting bone using waterjet should exceeds 80 MPa [30].

Although pulsed and abrasive waterjets achieve higher machining efficiencies and better surface quality than pure waterjet [62,68], surface cracks, pits, and roughness are unavoidably induced in bone as shown in Fig. 14(d). On waterjet-processed bone cement surfaces, chipping, micro-pits, isolated voids, and cracks are also produced by waterjet [59]. These defects are formed by waterjet compression forces and act as stress concentration points to cause more crack propagations in the cement [59]. The cement fractures when the stresses exceeds its strength, which generates cement debris at surgical sites [59]. Thus, suction devices needs to be used to remove the debris from the surgical sites [59]. In spite of the improved cutting efficiencies using abrasive waterjet [146], the process may cause the thromboembolic effect during

bone cutting. Therefore, caution must be taken in abrasive selections with respect to biocompatibility, pharmacological safety, high molecular weight, and physiological osmotic pressure [148].

4.3. Process parameter effects on bone cutting performance

Waterjet processing of bone mainly involves cutting and drilling operations. The operation characteristics are summarized in Table 7.

The cutting process parameters for bone include waterjet pressure, impact angle, diameter, and medium, all of which have an impact on bone removal rates [30]. Cutting depths tend to increase with higher pressure and impact angle [151]. Pulsed and abrasive waterjets have been found to achieve greater cutting depths compared to pure waterjets [30]. Owing to interactions of abrasives and bone, better surface quality can also be obtained with abrasive waterjet [68]. Shakouri et al. [68] studied the abrasive effect on bone surface roughness and found that looser, more uniformly aggregated sugar particles produced a more favorable abrasive feed, resulting in better surface quality. Surface roughness also decreased with increased water pressures due to increased abrasive velocities and kinetic energy [68] and decreased traverse velocities due to increased collision times between the abrasives and bone surfaces [68]. Biskup et al. [67] also found that lower pressures and abrasive flow rates can lower a temperature rise and shorten the exposure time to high temperatures and higher traverse velocities can have the opposite effect. Therefore, controlling of transverse velocities and temperature rises are critical to surface quality. However, abrasives applied must be water-soluble and pharmacologically safe [69].

Hloch et al. [62] studied the cutting of bone cement through the utilization of continuous and pulsed waterjets. As shown in Fig. 15, pulsed waterjet achieved deeper traces and better surface quality compared to continuous waterjet. During the disintegration of bone cement, higher depths were achieved when using a circular orifice [62]. The cutting depths increased with higher pressures [150] and acoustic chamber lengths [59] but decreased with traverse velocities [59]. The cutting depths also increased with an increased stand-off distance, with the optimal value being 4 mm [59]. However, they started to decrease with further increased distances [59]. Additionally, 0.9 % saline yielded

Table 8
Effects of waterjet process parameters on cutting depths, surface quality and temperatures.

Parameter	Effect on cutting/ drilling depth	Effect on surface quality	Main effect on temperature
Pressure	(+) Positive [15,59,62,145]	(+) Positive [68]	(+) Positive [67]
Traverse speed	(–) Negative [59]	(–) Negative [68]	(–) Negative [67]
Angle(0–90°)	(+) Positive [30,151]	Not mentioned	Not mentioned
Jet collision duration	(+) Positive [15,145,146]	Not mentioned	Not mentioned
Abrasive(Y/N)	(+) Positive [30,68,151]	(+) Positive [30,68]	(+) Positive [67]

greater cutting depths than water., due to their difference fluid densities [150].

Bach et al. developed a prediction model for drilling depths and hole diameters in cortical bone and polymethyl methacrylate. The model took into consideration the influence of abrasive flow rate, pressure, and time [15]. In static drilling, waterjet pressure is the most important parameter because high pressures above a specific pressure limit can induce shock waves to crack the materials. In *in vitro* waterjet drilling of bone, interferences between incoming and outgoing jets reduce impact pressures and kinetic energy [145]. Drilling depths increase with the pressure and the drilling time [145] and decrease with increased bone volume fraction [146] while drilling diameters decrease with the hole depth [145]. The drilling depth prediction equations were nonlinearly related to waterjet pressure, time, and bone volume fraction [146].

As bone cutting with waterjet is related to the process parameters, Table 8 summarizes the influences of waterjet drilling/cutting parameters on cutting/drilling depths, surface quality, and temperature rises based on published experimental and numerical modeling results. Increased pressures and decreased traverse velocities increase cutting depths, which also has a paraboloid relation with stand-off distances. While abrasive and pulsed waterjets yield higher removal rates and surface quality than pure waterjet, decreased traverse velocity and increased pressure also improve surface quality.

4.4. Potential applications of waterjet in orthopedic surgeries

So far, waterjet technology has not been clinically utilized in orthopedic surgery. Some studies suggested that waterjet technology, with its non-thermal cutting, tissue selectivity, and lower reactive forces, may potentially serve as a new tool for microfracture treatment and screw fixation in orthopedic surgery [14]. In *in vitro* experimental investigation on waterjet drilling with a pressure of 50 MPa for 5 s to treat talar chondral defects of goat [14]. The treatment outcomes were similar to those achieved with traditional drilling methods [14]. This method required stable water pressure, water jet beam coherence, stable positioning of the nozzle head when jetting, and minimizing excessive fluid extravasation. Waterjet may also be used in prosthetic repair surgeries or osteotomy as a minimally invasive tool for removal of interface tissues around sterile loose hips or knee prostheses [152]. In the feasibility studies of total hip and knee replacements using waterjet, bone cement can be removed without damaging bone tissue at 10–12 MPa with nozzle of 0.2 mm [152]. A small nozzle is preferable in the tissue removal instrument, because less water needs to be evacuated from the

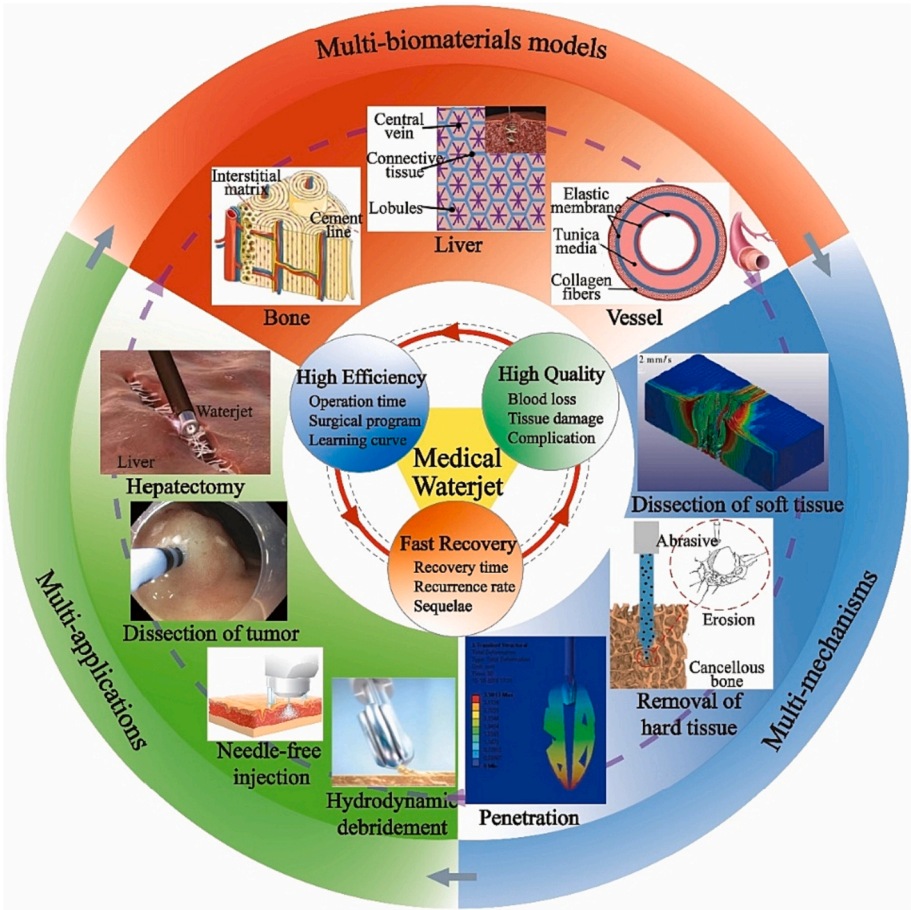


Fig. 16. Biomaterial models, removal mechanisms and clinical applications for waterjet processing in surgeries [5,7,25,71,73,112,139,153]. All figures reprinted with permissions from Elsevier, Springer, IEEE, Wiley and De Gruyter.

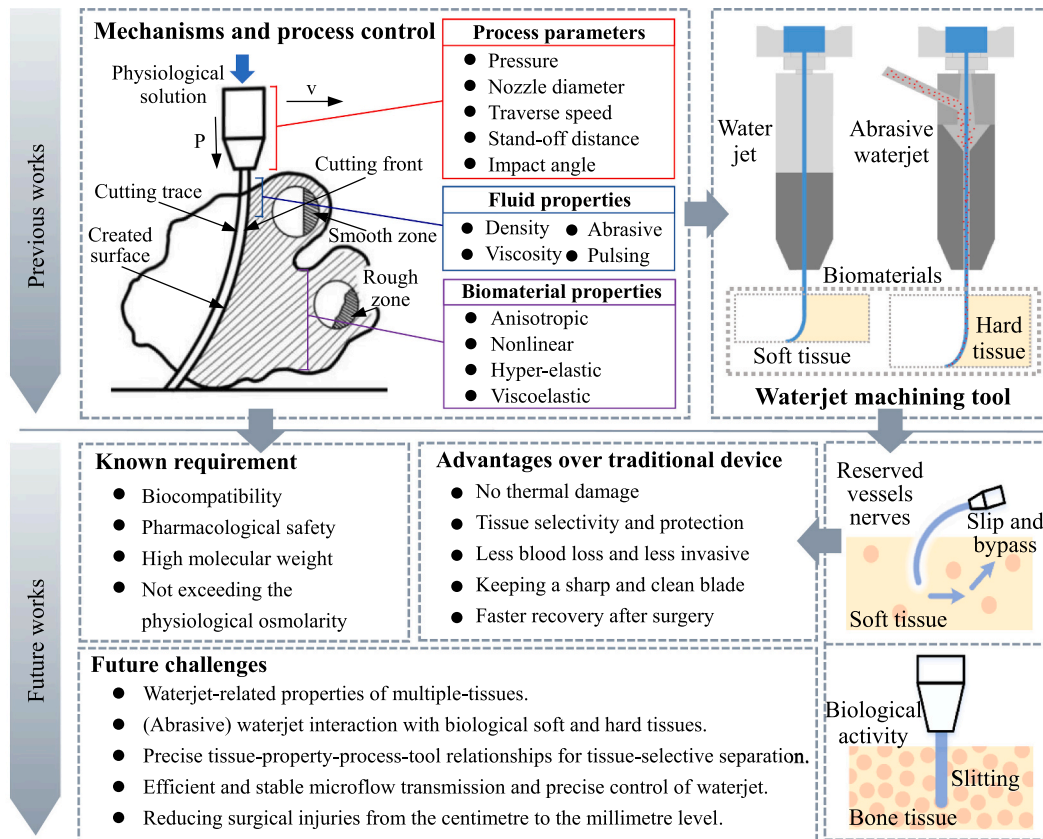


Fig. 17. Previous works and future challenges of waterjet techniques in surgeries.

periprosthetic cavity. It is important to note that the current research on waterjet technology for the above surgeries is still in its early stages. While initial animal/*in vitro* studies have shown promising results, further research and clinical trials will be necessary to fully understand the effectiveness, safety, and long-term outcomes of waterjet technique.

5. Conclusions and outlook

This paper conducted a comprehensive literature review on waterjet techniques in processing of tissues with multi-biomaterials models, multi-removal mechanisms, and multi-applications shown in Fig. 16. The fundamentals of waterjet processing of tissues include cutting characteristics, tissue separation mechanisms, surface formation mechanisms, tool solutions and clinical performance. Although waterjet techniques will be continuously expanded due to their advancement and versatility as surgical tools, there are still challenges for their applications. As shown in Fig. 17, the main limitations, challenges, and trends of waterjet techniques are drawn as follows:

5.1. Waterjet for soft tissues dissection

5.1.1. Tissue material properties

Although soft tissue structures and properties play key roles in waterjet dissection, existing research only focuses on limited tissue types. Multiple tissues properties associated with their material constitutive modes, physical and mechanical properties based on experimental measurements and investigations are missing. Efforts towards the understanding of such properties will facilitate the elucidation of tissue removal mechanisms and accelerate new tool development for waterjet dissection of soft tissues.

5.1.2. Tissue-waterjet interaction mechanisms

Due to the complex nature of waterjet dissection of soft tissues, little is known about the multiple tissue-waterjet interaction mechanisms. Studies considering failure mechanism and dynamic response process of waterjet dissection tissues using experimental investigation and models are expected. Particularly, the coupling material, mechanical and physical interactions between waterjet energy and tissue interfaces are critical but less to address. These missing microstructure-property-processing relations in waterjet cutting directly control crack propagation in soft tissues. The understanding of the tissue-process-performance relationships will provide scientific guidance to improve waterjet techniques for tissue dissection, especially for multiple-tissue-selective separation, which is clinically challenging and also future work.

5.1.3. Waterjet medium

Currently medical waterjet medium is almost saline, which is prone to be quickly absorbed by soft tissues, easily resulting in postoperative tissue adhesion, especially at closed tissue interfaces. Therefore, optimization and innovation of medical waterjet mediums considering both intraoperative cutting and postoperative repair effects are necessary for surgeries. Dissolution of polysaccharides with anti-adhesion functions (e.g., chitosan as a common anti-adhesion medical material) in saline might be promising for the development of novel waterjet mediums in the future.

5.1.4. Waterjet tool and process control

The stability of transmission and the control accuracy of waterjet pressures in waterjet processes are critical for surgeries. Effectively and accurately control transmission processes are paramount in tissue-selective separations given of the diversified nature of involved tissues. Advanced monitoring and data analysis algorithms may enable robot-assisted waterjet processes for soft tissues to achieve higher

precision and accuracy. Therefore, the combination of robots and waterjet should be further developed. The in-depth understanding and exploration of transmission theory and control methods for efficient and stable waterjet microflow transmissions and precise control will inspire further design and development of new tool structures.

5.1.5. Clinical applications and future trends

Although waterjet has been used for multiple surgical operations in open and laparoscopic surgeries, it will be continuously evolved for ultra-minimally invasive closed surgeries by combining image navigation technology. The new technique may provide a new solution for dissecting myofascial adhesion in closed surgery conditions with needle-level trauma for pain therapy. It will open a new research direction in waterjet technology transformation by reducing surgical scales from the centimeter to the millimeter level.

5.2. Waterjet for bone cutting

Despite of the advantage of waterjet for bone cutting, the mechanism of tissue-selective separation is only working in total hip and knee replacement operations for bone cement removal. As applied waterjet pressures for bone tissue removal is significantly higher than those for peripheral soft tissues, excessive damage in soft tissues often occurs. How to precisely control bone cutting to avoid peripheral soft-tissue injury in orthopedic surgeries is the key problem to be solved in the future. This requires a deep understanding for effects of abrasive type, size, and shape on dynamic characteristics and removal mechanisms of abrasive waterjet in bone cutting. More efforts should be made to explore tissue-property-process-performance relationships to make waterjet technique applicable in orthopedic surgeries.

5.3. Concluding remarks

Our knowledge of waterjet machining techniques and their applications in surgeries has improved significantly over the recent years. The challenges for enhancing the waterjet machining capability are exciting and present great opportunities for mechanical engineers, materials scientists, and surgeons to work together for future technical innovations and advances.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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